CODE APEX

THE END IS COMING.

POWERED BY verzii

Team Details

- Team name - **Mac_Enthu**
- Team leader Name - **Aayush Aade**
- Problem Statement (Domain) - **Track 1 (Emerson) · PS 1.2 Project IO SIGNAL SUDOKU**

SlotSmith · rule-driven, independently checked, revision-aware DeltaV I/O allocation

Problem Statement & Target Need

Who is affected

DeltaV project and I/O engineers (Emerson and system integrators) who must give every field signal an exact home: cabinet → CIOC → baseplate → CHARM slot. Downstream, panel builders and site teams wire to that allocation.

The current gap

Emerson's PS: 10,000+ signals are allocated by hand in Excel, and each customer revision means reworking it and tracing the impact again. DeltaV tools automate the steps around it (auto-sense, late binding, Bulk Edit); we found no public tool that places signals from project rules (internal tools may exist).

Why it matters now

Electronic Marshalling makes late changes physically cheap ("without lifting a wire"), **but allocation planning and revision tracking remain a manual workflow burden (per the PS).** Six interacting rules must hold at once, on every revision, with a technical query (TQ) for each data conflict.

Sources: Emerson PS #02 p.4; DeltaV Electronic Marshalling PDS (Aug 2026); Smart Commissioning WP (2021) p.5; Bulk Edit WP (2022).

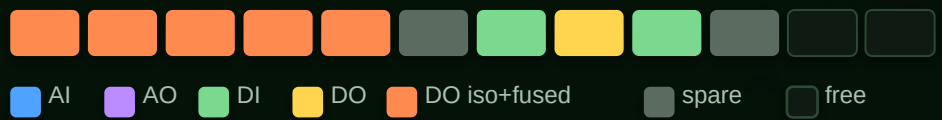
THE PUZZLE, PER EMERSON'S RULEBOOK

Cabinet ≤ 4 CIOCs → **CIOC** 8 baseplates → **Baseplate** 12 CHARMs → **CHARM** 1 signal

24-pair cable = 2 full baseplates (22 signals + 2 wired spares)



Smaller cables share a baseplate, in pair order



1 Keep IO groups together

2 A and B never share a CIOC

3 Capacity 12 · 8 · 4

4 Cable intact, in pair order

5 ~10% spare per cabinet

6 Right CHARM + terminal block

Proposed Solution & Innovation

What makes it different

- **Independently checked:** a separate verifier re-reads the exported workbook (11 rule families).
- **Revision-aware:** unchanged cables stay put; CP-SAT repairs changes locally.
- **Configurable:** existing rule parameters live in JSON; new rule types may need code.
- **Traceable TQs:** every TQ cites a rule ID and the MDB file and row.

Core mechanism

Each cable becomes one block of 24/12/6/2 slots (spares included). Blocks are packed per group and A/B side into 12-slot baseplates, then baseplates fill CIOCs (8) and cabinets (4, minus the spare reserve) in group order. **Proved:** for one A/B side with cable sizes 2/6/12 (24 = two full baseplates), first-fit-decreasing uses the fewest baseplates. **Tested, not proved:** whole-group totals matched an exact CP-SAT solver in 19,682 exhaustive small cases and 12/12 benchmark runs.

Why it's feasible in this timeframe

Built on Python, OR-Tools CP-SAT and openpyxl, all open source, with a synthetic MDB generator built from the published rulebook. The working prototype already allocates and verifies 20,033 synthetic signals in 6.0 s on a laptop (slide 6).

WHERE AI HELPS, AND WHERE IT MUST NOT

AI-assisted, always reviewed

- Column auto-mapping for unseen MDBs
- Plain-English rule → JSON draft, schema-checked and previewed
- Customer-ready TQ wording

Status: planned for Round 2+

Solver + independent verifier

- Every placement decision
- Rule verification
- Excel export and revision diff

Status: built and tested

DESIGN PRINCIPLES

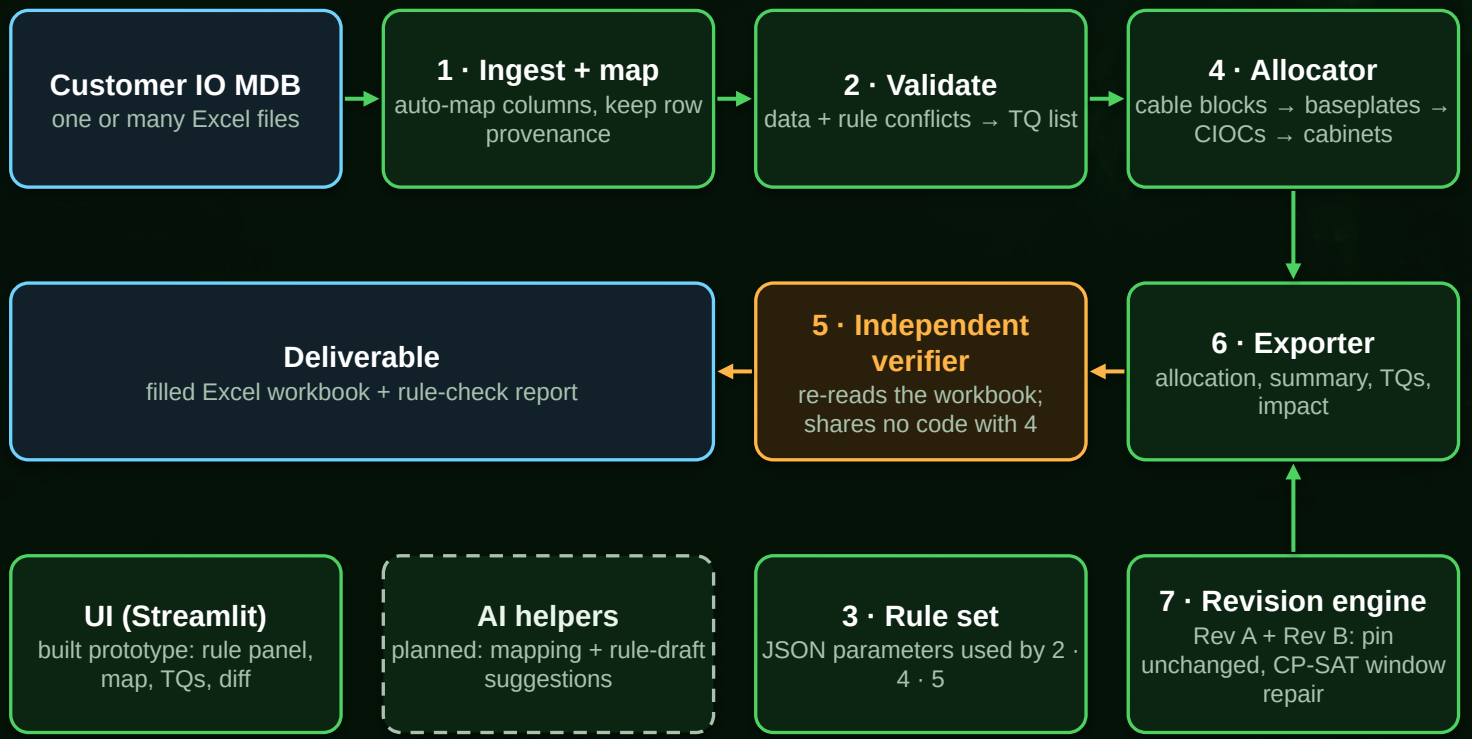
- **Check, don't trust:** nothing ships without an independent rule-check report
- **No AI in placement:** a solver places signals; AI only suggests, users confirm
- **Change locally:** a revision keeps unchanged cables in place and reports every move
- **Surface, don't hide:** every unresolved issue becomes a TQ with its rule and source row

System Architecture & Tech Stack

TECH STACK

Frontend	Streamlit prototype (built): rule panel, cabinet map, TQs, diff; React + TS later
Backend	Python 3.12 · pandas · openpyxl · pydantic typed rule sets · CLI
Database	Project files: JSON rule sets, Rev A/B workbooks, run log (SQLite planned)
AI / APIs	OR-Tools CP-SAT (Apache-2.0). Optional LLM for mapping/rule drafts: model declared, never places signals
Hosting / DevOps	Runs offline on a laptop · pytest + Hypothesis · git; Docker for demo (planned)

DATA FLOW (solid = built; dashed = planned)



Row provenance (file:sheet:row) travels with every signal, so each TQ and placement is traceable.

Key Features & UI/UX

1 · Upload
Excel MDB(s) in; columns auto-mapped with provenance

Built

2 · Set rules
Rule panel: toggle, edit, save rule sets per project

Built (prototype panel)

3 · Allocate
One click: signal → cabinet, CIOC, baseplate, CHARM

Built

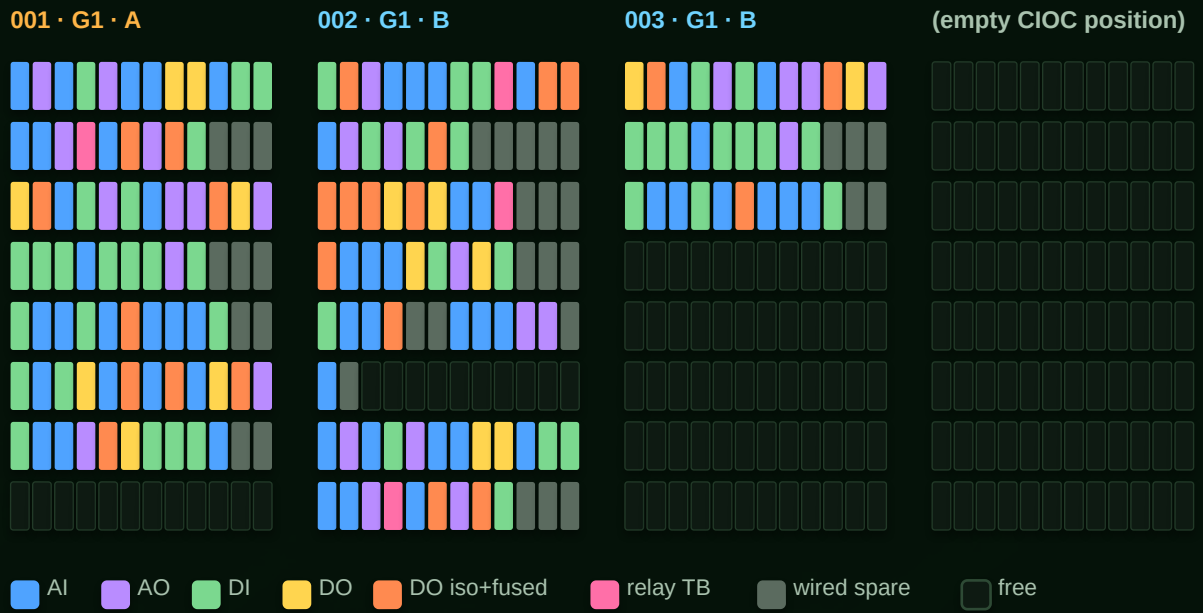
4 · Report + TQs
Fill %, spares, rule check, TQ list with sources

Built

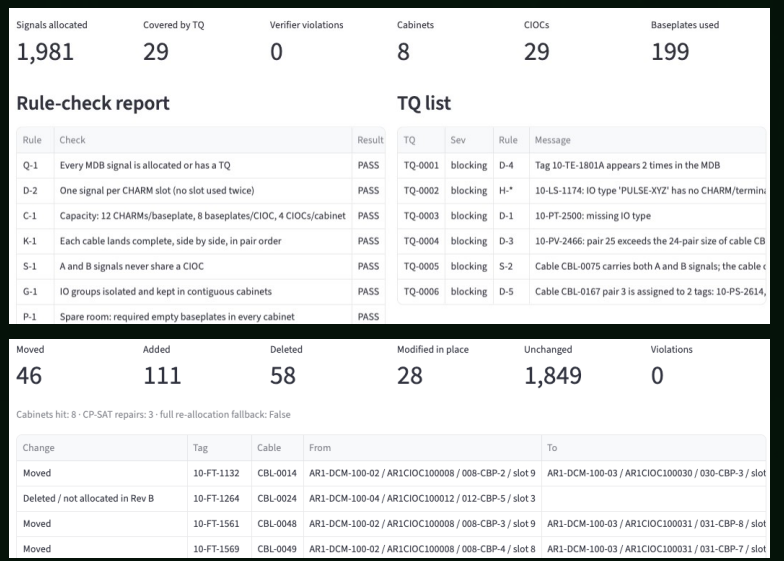
5 · Revise
Rev B vs Rev A: unchanged stay put; impact report

Built

ENGINE OUTPUT: CABINET AR1-DCM-100-01 *(synthetic 900-signal MDB)*



PROTOTYPE UI (real screenshots, post-audit build)



Signals allocated	Covered by TQ	Verifier violations	Cabinets	CIOCs	Baseplates used
1,981	29	0	8	29	199

Rule-check report

Rule	Check	Result
Q-1	Every MDB signal is allocated or has a TQ	PASS
D-2	One signal per CHARM slot (no slot used twice)	PASS
C-1	Capacity: 12 CHARMs/baseplate, 8 baseplates/CIOC, 4 CIOCs/cabinet	PASS
K-1	Each cable lands complete, side by side, in pair order	PASS
S-1	A and B signals never share a CIOC	PASS
G-1	IO groups isolated and kept in contiguous cabinets	PASS
P-1	Spare room: required empty baseplates in every cabinet	PASS

TQ list

TQ	Sev	Rule	Message
TQ-0001	blocking	D-4	Tag 10-TE-1801A appears 2 times in the MDB
TQ-0002	blocking	H-*	10-LS-1174: IO type 'PULSE-XYZ' has no CHARM/termina
TQ-0003	blocking	D-1	10-PT-2500: missing IO type
TQ-0004	blocking	D-3	10-PV-2466: pair 25 exceeds the 24-pair size of cable CB
TQ-0005	blocking	S-2	Cable CBL-0075 carries both A and B signals; the cable c
TQ-0006	blocking	D-5	Cable CBL-0167 pair 3 is assigned to 2 tags: 10-PS-2614,

Moved	Added	Deleted	Modified in place	Unchanged	Violations
46	111	58	28	1,849	0

Cabinets hit: 8 · CP-SAT repairs: 3 · full re-allocation fallback: False

Change	Tag	Cable	From	To
Moved	10-FT-1132	CBL-0014	AR1-DCM-100-02 / AR1CIOC100008 / 008-CBP-2 / slot 9	AR1-DCM-100-03 / AR1CIOC100030 / 030-CBP-3 / slot
Deleted / not allocated in Rev B	10-FT-1264	CBL-0024	AR1-DCM-100-04 / AR1CIOC100012 / 012-CBP-5 / slot 3	
Moved	10-FT-1561	CBL-0048	AR1-DCM-100-02 / AR1CIOC100008 / 008-CBP-3 / slot 9	AR1-DCM-100-03 / AR1CIOC100031 / 031-CBP-8 / slot
Moved	10-FT-1569	CBL-0049	AR1-DCM-100-02 / AR1CIOC100008 / 008-CBP-4 / slot 8	AR1-DCM-100-03 / AR1CIOC100031 / 031-CBP-7 / slot

SYNTHETIC 2,000-signal MDB (6 injected errors) + synthetic Rev B. Top: allocate. Bottom: revise.

Implementation Roadmap & Scope

What's built today, what's in progress, and what's planned beyond the hackathon

BUILT TODAY · 10 OCT

- Synthetic MDB + Rev B generator (seeded)
- Ingestion: multi-file, header detection, auto-map
- Validation → TQs citing rule + source row
- Allocator: cable blocks, FFD, groups, A/B, spares, CHARM/TB, names
- Independent verifier: 11 rule families
- Excel export incl. rule check + impact
- Revision engine + CP-SAT window repair
- Streamlit prototype UI
- 31 automated tests incl. property-based

ROUND 2 · BY 16 OCT

- UI polish: mapping review/override, CHARM-table editing, exports
- Map to the organiser template once released
- Confirm open rule interpretations with mentors
- Cut CP-SAT repair time and run-to-run variance

GRAND FINALE · 24 OCT

- Run on organiser samples + hidden test
- AI helpers: column-mapping suggestions, plain-English → rule JSON (previewed)
- Demo hardening, offline bundle

BEYOND THE HACKATHON

- Bulk-Edit-style export for DeltaV import (format to be validated)
- Other electronic-marshalling platforms (untested)
- Cabinet labels and drawings

MEASURED ON SYNTHETIC DATA (prototype, Apple M5) · *not evidence about the hidden test*

0
verifier violations in 12 runs (500–20k signals)

100%
of rows allocated or covered by a TQ

81/81
injected data errors raised as TQs

0
baseplates above the exact per-group minimum (12 runs)

5.99 s
end-to-end for 20,033 signals (median)

429–470 vs 49,805
moved: SlotSmith vs re-run, 10 Rev B cases, 3 runs; CP-SAT proven exact in 7/11 window repairs

Expected Impact & Scalability

Who benefits, how big the opportunity is, and how this grows beyond a demo.

IO / project engineers

Allocation becomes a re-runnable, rule-checked step instead of hand-editing 10,000+ Excel rows.

Reviewers and QA

Every deliverable ships with a rule-check report; every placement and TQ traces to an MDB row.

Customers and EPCs

Specific, traceable TQs and a Rev A → Rev B impact report: added, deleted, moved, cabinets hit.

HOW BIG THE OPPORTUNITY IS (sourced)

10,000+	signals per project, allocated by hand today (Emerson PS)
≤ 300	CIOCs per DeltaV system (spec maximum) → at most 28,800 CHARM channels (EM PDS)
Every	customer revision means rework and impact tracing (Emerson PS)

HOW IT GROWS BEYOND A DEMO

- Rule sets saved per customer and project: the next project starts from the last one
- Designed to complement DeltaV and SPI: an export adapter to their import formats is planned (not yet validated)
- Block-packing engine could be adapted to other electronic-marshalling I/O families (untested)
- Near-linear runtime measured: 0.6 s at 2k, 3.0 s at 10k, 6.0 s at 20k signals (synthetic)

HONEST LIMITS TODAY

Synthetic data only: the organiser's sample MDB and template are not yet released, so no official-template compatibility is claimed. Rule interpretations (straddling, spare rounding, controller mapping) await mentors. Rules are parameters; new rule types still need code. CP-SAT repair is window-local. No time savings claimed. Complements DeltaV tools; does not replace them.

CODE APEX

BUILD BEFORE IT DOES.

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THANK YOU

Team **Mac_Enthu** · Aayush Aade (lead) · Siddhant Patil · Gaurang Bagwe